

EDUCATION

- **University of South Dakota** Vermillion, SD
Master of Science in Computer Science (AI Specialization) Aug. 2023 - May 2025
- **Jawaharlal Nehru Technological University** Hyderabad, India
Bachelor of Technology in Computer Science Engineering Aug. 2017 - July 2021

TECHNICAL SKILLS

- **Responsible AI & Social Computing:** Model Governance, Technological Harm Reparation, Alignment Vulnerability Analysis, Sensemaking Evaluation Frameworks
- **Core Languages & Full-Stack UI:** Python, C#, Java, Go (Golang), React, JS, Core OOP
- **Enterprise Backend & Architecture:** PostgreSQL, Azure, Docker, RESTful APIs, gRPC, Protocol Buffers, ETL Data Workflows

PROFESSIONAL EXPERIENCE

- **University of South Dakota (Applied AI Lab)** Vermillion, SD
Graduate Research Assistant Aug. 2023 - May 2025
 - Curated a 105,000-sentence enterprise data workflow addressing the reparation of technological harm for affected communities, earning the Best Paper Award (RTIP2P-2024) for language preservation.
 - Integrated Responsible AI across the innovation pipeline by architecting community support protocols and facilitating peer research for the University of Washington / Wisconsin Undergraduate Honors Symposium.
 - Executed systematic data poisoning and prompt injection campaigns to identify alignment vulnerabilities in production-level models, ensuring robust model governance and safe social computing integration.
 - Administered concurrent job schedulers and containerized environments on high-performance supercomputing clusters, maintaining strict scientific reproducibility for algorithmic architectures.
- **W. H. Over Museum** Vermillion, SD
Systems Engineer (AI & Automation) July 2025 - Present
 - Designed client-facing RESTful API endpoints integrating interactive 3D models with the core CMS, directly supporting advanced Human-Computer Interaction (HCI) paradigms for virtual exhibits.
 - Engineered programmatic textual unit tests to sanitize flawed legacy data, ensuring high-fidelity sensemaking capabilities across unified and protected database schemas.
 - Architected a zero-API-cost, privacy-first ETL data migration pipeline, resulting in the successful transfer of 61,262 records to a serverless PostgreSQL cloud database hosted on Microsoft Azure.
 - Implemented strict Role-Based Access Controls and automated checksum validation scripts to guarantee 100.00% metadata accuracy and systemic protection.
- **Independent Consultant** Hyderabad, India
Software Developer Aug. 2021 - July 2023
 - Architected a highly scalable distributed master-worker network utilizing gRPC and dynamic task reassignment protocols, achieving 99.50% availability during simulated outages.
 - Developed a modular engine implementing custom natural language string parsers and constant-time memory lookup efficiency, preventing logic injection errors.

TECHNICAL PROJECTS

- **LakotaBERT: Responsible AI & Social Computing Impact:** Engineered a Transformer-based NLP architecture tailored for affected communities to prevent language extinction. Achieved Best Paper Award recognition and featured in the PWNA 2025 Year in Review and AITRAN "Voices Unextinguished" for mitigating technological harm.
- **Academic Search & Sensemaking Platform:** Developed a 10-point algorithmic evaluation framework utilizing Time-Bound Mean Reciprocal Rank to quantify database comprehensiveness, metadata integrity, and true retrieval efficacy for human sensemaking.
- **LLM Vulnerability & Alignment Evaluation:** Conducted dual-phase white-box and black-box security testing against foundation models to parse out threat vectors. Quantified how adversarial data manipulates alignment and inverts factual classifications.

CERTIFICATIONS & MEMBERSHIPS

- **IEEE Computer Society:** Certified Software Professional (Level 2)
- **IEEE:** Designing Security Solutions for Edge, Cloud, and IoT
- **IEEE:** Active Member (6 Years) - Workshops on AI and Systems Engineering